Risk Management

Professor Philip Howard
howardpd@wfu.edu

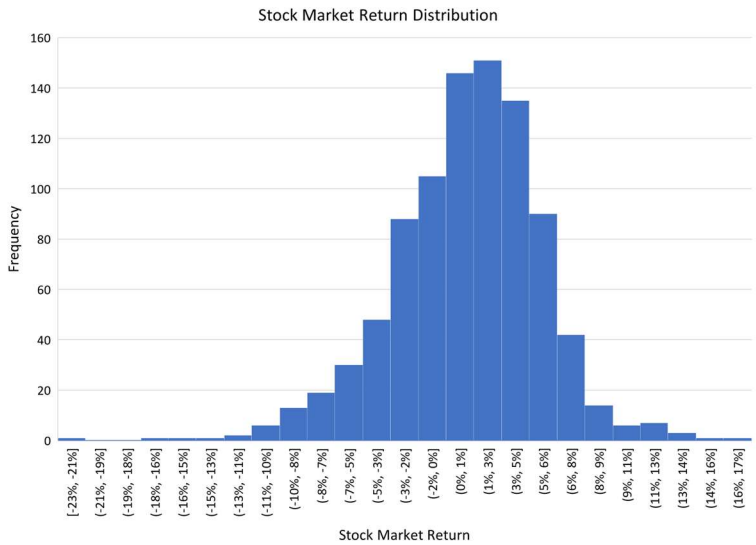
WAKE FOREST
UNIVERSITY
SCHOOL of BUSINESS

Value-at-Risk

Professor Philip Howard
howardpd@wfu.edu

WAKE FOREST
UNIVERSITY
SCHOOL of BUSINESS

Historical Market Returns



Normal Distribution & Finance

- Common to assume returns are normally distributed
  - Mean is a measure of reward
  - Standard deviation is a measure of risk
  - No theoretical justification for assumption
  - Empirically, Normality is close and is a good starting point for financial models
  - However, in the data we do observe
    - Negative skewness
    - Positive kurtosis
  - This implies we underestimate tail risk in our financial models
  - To the extent returns are not normally distributed
    - Standard deviation is no longer a complete measure of risk
    - Sharpe ratio is not a complete measure of portfolio performance

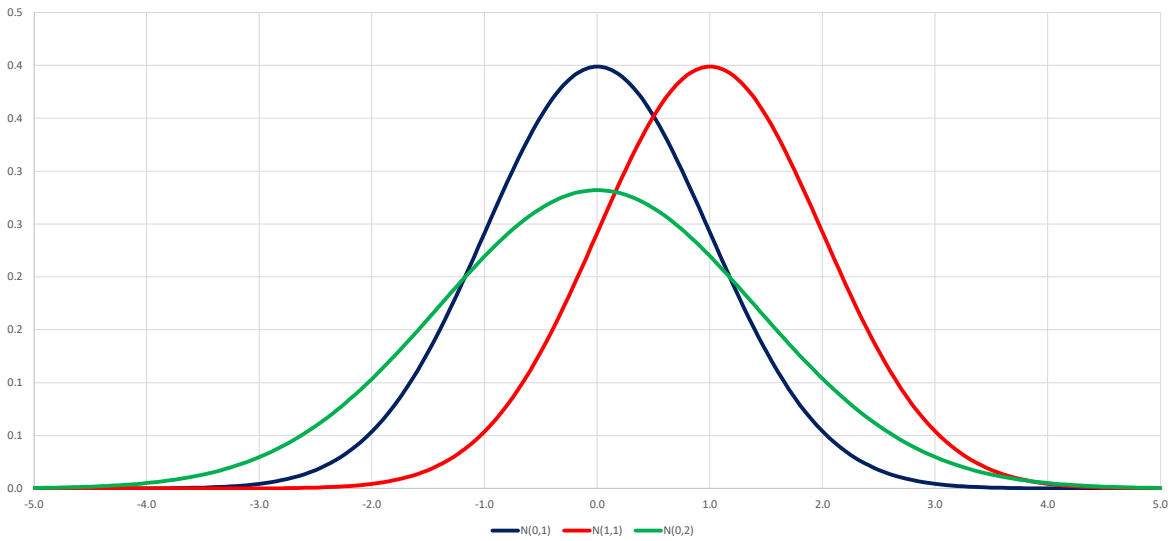
Random Variables

- Random variable has random outcomes
- Outcomes have probabilities
- Two types of probability distributions
  - Discrete: outcomes are discrete values
  - Continuous: outcomes over continuous interval
  - Probability distributions sum to 1

Normal Distribution

- Normal: $x \sim N(\mu, \sigma^2)$
  - Shape of distribution is bell curve
  - Mean of distribution is μ
  - Variance of distribution is σ^2

Normal Distribution $x \sim N(\mu, \sigma^2)$



Normal Random Variable

- Probability of $-\infty \leq x \leq a$ is the area under curve from $-\infty$ to a

$$P(-\infty \leq x \leq a) = \text{norm.dist}(a, \mu, \sigma, \text{TRUE})$$

- Probability of a $\leq x \leq b$ is the area under curve from a to b

$$P(a \leq x \leq b) = P(-\infty \leq x \leq b) - P(-\infty \leq x \leq a)$$

- Probability of $a \leq x \leq \infty$ is the area under curve from a to ∞

$$P(a \leq x \leq \infty) = 1 - P(-\infty \leq x \leq a)$$

Normal Random Variable

- Probability of $-\infty \leq x \leq a$ is the area under curve from $-\infty$ to a

$$P(-\infty \leq x \leq a) = \text{norm.dist}(a, \mu, \sigma, \text{TRUE})$$

- The value on the x-axis, a, that corresponds to probability $p = P(-\infty \leq x \leq a)$

$$a = \text{norm.inv}(p, \mu, \sigma)$$

Example

- $\mu = 1$
- $\sigma = 3$

- $P(-\infty \leq x \leq 5) = \text{norm.dist}(5,1,3, \text{TRUE}) = 91\%$
- a such that corresponds to probability $91\% = P(-\infty \leq x \leq a)$

$$a = \text{norm.inv}(91\%, 1,3) = 5$$

Risk Analysis & Value at Risk (VaR)

- p VaR – Loss such that there is a probability p of observing at least as extreme

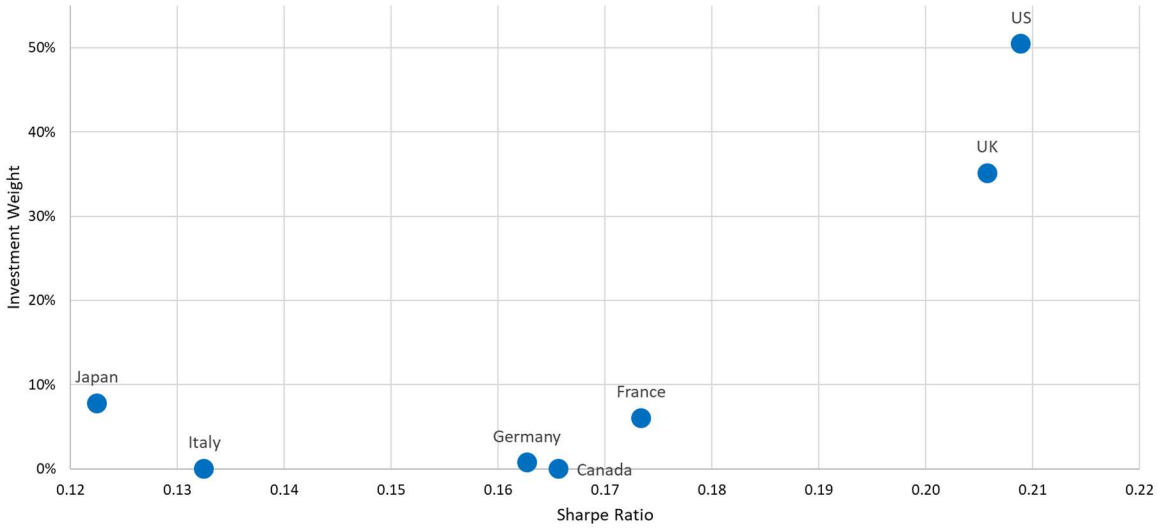
- If asset has 5% VaR of $1 million, there is a 5% probability of the asset losing $1 million or more

International Diversification

Professor Philip Howard
howardpd@wfu.edu

WAKE FOREST
UNIVERSITY
SCHOOL of BUSINESS

International Diversification



Statistical Arbitrage

Professor Philip Howard
howardpd@wfu.edu

WAKE FOREST
UNIVERSITY
SCHOOL of BUSINESS

Linear Combinations of Random Variables

Let a, b and c be constants and X, Y and Z be random variables

- $E[aX + bY] = aE[X] + bE[Y]$
- $Var(aX + bY) = a^2Var(X) + b^2Var(Y) + 2ab \cdot Cov(X, Y)$
- $Cov(aX + bY, cZ) = ac \cdot Cov(X, Z) + bc \cdot Cov(Y, Z)$

- Note: $Cov(X, X) \equiv Var(X)$

CAPM

$$R_i = R_f + \beta_i(R_M - R_f)$$

- $\beta_i = \frac{\sigma_{iM}}{\sigma_M^2}$: Asset i's beta

CAPM Portfolio Betas

CAPM portfolio betas are additive

$$
\begin{aligned}
\beta_P &= \frac{\sigma_{PM}}{\sigma_M^2} \\
&= \frac{Cov(R_P, R_M)}{\sigma_M^2} \\
&= \frac{Cov(w_A R_A + w_G R_G, R_M)}{\sigma_M^2} \\
&= \frac{w_A \cdot Cov(R_A, R_M) + w_G \cdot Cov(R_G, R_M)}{\sigma_M^2} \\
&= w_A \frac{\sigma_{AM}}{\sigma_M^2} + w_G \frac{\sigma_{GM}}{\sigma_M^2} \\
&= w_A \beta_A + w_G \beta_G
\end{aligned}
$$

Market Neural Portfolios

- A popular hedge fund strategy is “pairs trading” or “statistical arbitrage”
- Create a portfolio consisting of two assets that has 0 market beta
- The portfolio has no market risk, and purely speculating on the relative performance of the two assets